

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
df=pd.read_csv('/content/mtcars2.csv')
```

```
print(df)
```

	S.No	Unnamed: 1	mpg	cyl	disp	hp	drat	wt	qsec	vs	\
0	1	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	
1	2	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	
2	3	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	
3	4	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	
4	5	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	
5	6	Valiant	18.1	6	225.0	105	2.76	3.460	NaN	1	
6	7	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	
7	8	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	
8	9	Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	
9	10	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	
10	11	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	
11	12	Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	
12	13	Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	
13	14	Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	
14	15	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	
15	16	Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	
16	17	Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	
17	18	Fiat 128	32.4	4	78.7	66	4.08	2.200	NaN	1	
18	19	Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	
19	20	Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	
20	21	Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	
21	22	Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	
22	23	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	
23	24	Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	
24	25	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	
25	26	Fiat X1-9	27.3	4	79.0	66	4.08	1.935	NaN	1	
26	27	Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	
27	28	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	
28	29	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	
29	30	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	
30	31	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	
31	32	Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	

	am	gear	carb
0	1	4	4
1	1	4	4
2	1	4	1
3	0	3	1
4	0	3	2
5	0	3	1
6	0	3	4
7	0	4	2
8	0	4	2
9	0	4	4
10	0	4	4
11	0	3	3
12	0	3	3
13	0	3	3
14	0	3	4
15	0	3	4
16	0	3	4
17	1	4	1
18	1	4	2
19	1	4	1
20	0	3	1
21	0	3	2
22	0	3	2

```
df['mpg']=df['mpg'].astype(str)
```

```
df['mpg'],df['cyl'],df['disp'],df['hp'],df['wt'],df['qsec'],df['vs'],df['cyl'],df['am'],df['gear'],df['carb']).dtypes
```

```
(0    21.0
1    21.0
2    22.8
3    21.4
4    18.7
```

```
5 18.1
6 14.3
7 24.4
8 22.8
9 19.2
10 17.8
11 16.4
12 17.3
13 15.2
14 10.4
15 10.4
16 14.7
17 32.4
18 30.4
19 33.9
20 21.5
21 15.5
22 15.2
23 13.3
24 19.2
25 27.3
26 26.0
27 30.4
28 15.8
29 19.7
30 15.0
31 21.4
Name: mpg, dtype: object,
0 6
1 6
2 4
3 6
4 8
5 6
6 8
7 4
8 4
9 6
10 6
11 8
12 8
13 8
14 8
15 8
16 8
17 4
18 4
19 4
20 4
21 8
22 8
23 8
```

df.head(5)

	S.No	Unnamed: 1	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	1	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	2	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
2	3	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
3	4	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	5	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2

df.head(10)

	S.No	Unnamed: 1	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	1	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	2	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
2	3	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
3	4	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	5	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	6	Valiant	18.1	6	225.0	105	2.76	3.460	NaN	1	0	3	1
6	7	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	8	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
8	9	Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
9	10	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4

df.tail(5)

	S.No	Unnamed: 1	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
27	28	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.9	1	1	5	2
28	29	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.5	0	1	5	4
29	30	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.5	0	1	5	6
30	31	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.6	0	1	5	8
31	32	Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.6	1	1	4	2

df.tail(10)

	S.No	Unnamed: 1	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
22	23	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	0	3	2
23	24	Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	0	3	4
24	25	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	0	3	2
25	26	Fiat X1-9	27.3	4	79.0	66	4.08	1.935	NaN	1	1	4	1
26	27	Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
27	28	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
28	29	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4
29	30	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6
30	31	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
31	32	Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2

df.shape

(32, 13)

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32 entries, 0 to 31
Data columns (total 13 columns):
#   Column      Non-Null Count  Dtype
---  -
0   S.No        32 non-null    int64
1   Unnamed: 1  32 non-null    object
2   mpg         32 non-null    object
3   cyl         32 non-null    int64
4   disp        32 non-null    float64
5   hp          32 non-null    int64
6   drat        32 non-null    float64
7   wt          32 non-null    float64
8   qsec        29 non-null    float64
9   vs          32 non-null    int64
10  am          32 non-null    int64
```

```
11 gear      32 non-null    int64
12 carb      32 non-null    int64
dtypes: float64(4), int64(7), object(2)
memory usage: 3.4+ KB
```

```
df.mean(numeric_only=True)
```

	0
S.No	16.500000
cyl	6.187500
disp	230.721875
hp	146.687500
drat	3.596563
wt	3.217250
qsec	17.674828
vs	0.437500
am	0.406250
gear	3.687500
carb	2.812500

dtype: float64

```
df.median(numeric_only=True)
```

	0
S.No	16.500
cyl	6.000
disp	196.300
hp	123.000
drat	3.695
wt	3.325
qsec	17.420
vs	0.000
am	0.000
gear	4.000
carb	2.000

dtype: float64

```
df.std(numeric_only=True)
```

```
0
```

	0
S.No	9.380832
cyl	1.785922
disp	123.938694
hp	68.562868
drat	0.534679
wt	0.978457
qsec	1.780394
vs	0.504016
am	0.498991
gear	0.737804
carb	1.615200

dtype: float64

```
df.max(numeric_only=True)
```

```
0
```

	0
S.No	32.000
cyl	8.000
disp	472.000
hp	335.000
drat	4.930
wt	5.424
qsec	22.900
vs	1.000
am	1.000
gear	5.000
carb	8.000

dtype: float64

```
df.min(numeric_only=True)
```

```
0
```

	0
S.No	1.000
cyl	4.000
disp	71.100
hp	52.000
drat	2.760
wt	1.513
qsec	14.500
vs	0.000
am	0.000
gear	3.000
carb	1.000

dtype: float64

```
df.isna().sum()#to see null values in data
```

	0
S.No	0
Unnamed: 1	0
mpg	0
cyl	0
disp	0
hp	0
drat	0
wt	0
qsec	3
vs	0
am	0
gear	0
carb	0

dtype: int64

df.notna().sum()#to see not null values in data

	0
S.No	32
Unnamed: 1	32
mpg	32
cyl	32
disp	32
hp	32
drat	32
wt	32
qsec	29
vs	32
am	32
gear	32
carb	32

dtype: int64

df.describe()

	S.No	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
count	32.000000	32.000000	32.000000	32.000000	32.000000	32.000000	29.000000	32.000000	32.000000	32.000000	32.0000
mean	16.500000	6.187500	230.721875	146.687500	3.596563	3.217250	17.674828	0.437500	0.406250	3.687500	2.8125
std	9.380832	1.785922	123.938694	68.562868	0.534679	0.978457	1.780394	0.504016	0.498991	0.737804	1.6152
min	1.000000	4.000000	71.100000	52.000000	2.760000	1.513000	14.500000	0.000000	0.000000	3.000000	1.0000
25%	8.750000	4.000000	120.825000	96.500000	3.080000	2.581250	16.870000	0.000000	0.000000	3.000000	2.0000
50%	16.500000	6.000000	196.300000	123.000000	3.695000	3.325000	17.420000	0.000000	0.000000	4.000000	2.0000
75%	24.250000	8.000000	326.000000	180.000000	3.920000	3.610000	18.600000	1.000000	1.000000	4.000000	4.0000
max	32.000000	8.000000	472.000000	335.000000	4.930000	5.424000	22.900000	1.000000	1.000000	5.000000	8.0000

df = df.rename(columns={'Unnamed: 1': 'model'})
print(df)

	S.No	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	\
0	1	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	
1	2	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	
2	3	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	
3	4	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	
4	5	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	
5	6	Valiant	18.1	6	225.0	105	2.76	3.460	NaN	1	
6	7	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	
7	8	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	
8	9	Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	
9	10	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	
10	11	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	
11	12	Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	
12	13	Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	
13	14	Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	
14	15	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	
15	16	Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	
16	17	Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	
17	18	Fiat 128	32.4	4	78.7	66	4.08	2.200	NaN	1	
18	19	Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	
19	20	Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	
20	21	Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	
21	22	Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	
22	23	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	
23	24	Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	
24	25	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	
25	26	Fiat X1-9	27.3	4	79.0	66	4.08	1.935	NaN	1	
26	27	Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	
27	28	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	
28	29	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	
29	30	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	
30	31	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	
31	32	Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	

	am	gear	carb
0	1	4	4
1	1	4	4
2	1	4	1
3	0	3	1
4	0	3	2
5	0	3	1
6	0	3	4
7	0	4	2
8	0	4	2
9	0	4	4
10	0	4	4
11	0	3	3
12	0	3	3
13	0	3	3
14	0	3	4
15	0	3	4
16	0	3	4
17	1	4	1
18	1	4	2
19	1	4	1
20	0	3	1
21	0	3	2
22	0	3	2

```
df['qsec'] = df['qsec'].fillna(df['qsec'].mean())
print(df['qsec'])
```

```
0    16.460000
1    17.020000
2    18.610000
3    19.440000
4    17.020000
5    17.674828
6    15.840000
7    20.000000
8    22.900000
9    18.300000
10   18.900000
11   17.400000
12   17.600000
13   18.000000
14   17.980000
15   17.820000
16   17.420000
17   17.674828
18   18.520000
19   19.900000
20   20.010000
```

```

21 16.870000
22 17.300000
23 15.410000
24 17.050000
25 17.674828
26 16.700000
27 16.900000
28 14.500000
29 15.500000
30 14.600000
31 18.600000
Name: qsec, dtype: float64

```

```

df1=df.drop(columns=['S.No'])
df1

```

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.460000	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.020000	0	1	4	4
2	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.610000	1	1	4	1
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.440000	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.020000	0	0	3	2
5	Valiant	18.1	6	225.0	105	2.76	3.460	17.674828	1	0	3	1
6	Duster 360	14.3	8	360.0	245	3.21	3.570	15.840000	0	0	3	4
7	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.000000	1	0	4	2
8	Merc 230	22.8	4	140.8	95	3.92	3.150	22.900000	1	0	4	2
9	Merc 280	19.2	6	167.6	123	3.92	3.440	18.300000	1	0	4	4
10	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.900000	1	0	4	4
11	Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.400000	0	0	3	3
12	Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.600000	0	0	3	3
13	Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.000000	0	0	3	3
14	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.980000	0	0	3	4
15	Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.820000	0	0	3	4
16	Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.420000	0	0	3	4
17	Fiat 128	32.4	4	78.7	66	4.08	2.200	17.674828	1	1	4	1
18	Honda Civic	30.4	4	75.7	52	4.93	1.615	18.520000	1	1	4	2
19	Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.900000	1	1	4	1
20	Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.010000	1	0	3	1
21	Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.870000	0	0	3	2
22	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.300000	0	0	3	2
23	Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.410000	0	0	3	4
24	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.050000	0	0	3	2
25	Fiat X1-9	27.3	4	79.0	66	4.08	1.935	17.674828	1	1	4	1
26	Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.700000	0	1	5	2
27	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.900000	1	1	5	2
28	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.500000	0	1	5	4
29	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.500000	0	1	5	6
30	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.600000	0	1	5	8

```

# Convert mpg to numeric type
df['mpg'] = pd.to_numeric(df['mpg'], errors='coerce')

```

```

# Calculate correlation matrix for numeric columns
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)

```



```

      S.No      mpg      cyl      disp      hp      drat      wt  \
S.No  1.000000  0.084699 -0.028882 -0.022048  0.242295  0.240952 -0.162419
mpg   0.084699  1.000000 -0.852162 -0.847551 -0.776168  0.681172 -0.867659
cyl   -0.028882 -0.852162  1.000000  0.902033  0.832447 -0.699938  0.782496
disp  -0.022048 -0.847551  0.902033  1.000000  0.790949 -0.710214  0.887980
hp     0.242295 -0.776168  0.832447  0.790949  1.000000 -0.448759  0.658748
drat   0.240952  0.681172 -0.699938 -0.710214 -0.448759  1.000000 -0.712441
wt     -0.162419 -0.867659  0.782496  0.887980  0.658748 -0.712441  1.000000
qsec  -0.362390  0.360351  0.548775 -0.385207 -0.650674  0.120175 -0.130362
vs     -0.109162  0.664039 -0.810812 -0.710416 -0.723097  0.440278 -0.554916
am     0.354905  0.599832 -0.522607 -0.591227 -0.243204  0.712711 -0.692495
gear   0.400825  0.480285 -0.492687 -0.555569 -0.125704  0.699610 -0.583287
carb   0.174576 -0.550925  0.526988  0.394977  0.749812 -0.090790  0.427606

      qsec      vs      am      gear      carb
S.No -0.362390 -0.109162  0.354905  0.400825  0.174576
mpg   0.360351  0.664039  0.599832  0.480285 -0.550925
cyl   -0.548775 -0.810812 -0.522607 -0.492687  0.526988
disp  -0.385207 -0.710416 -0.591227 -0.555569  0.394977
hp     -0.650674 -0.723097 -0.243204 -0.125704  0.749812
drat   0.120175  0.440278  0.712711  0.699610 -0.090790
wt     -0.130362 -0.554916 -0.692495 -0.583287  0.427606
qsec   1.000000  0.667873 -0.271763 -0.203784 -0.573987
vs     0.667873  1.000000  0.168345  0.206023 -0.569607
am     -0.271763  0.168345  1.000000  0.794059  0.057534
gear   -0.203784  0.206023  0.794059  1.000000  0.274073
carb   -0.573987 -0.569607  0.057534  0.274073  1.000000

```

```

#Fill any newly created NaN values in mpg
df['mpg'] = df['mpg'].fillna(df['mpg'].mean())

```

Double-click (or enter) to edit

```
df['mpg']=pd.to_numeric(df['mpg'],errors='coerce')
```

```
print(df.dtypes)
```

```

S.No      int64
model      object
mpg      float64
cyl      int64
disp      float64
hp      int64
drat      float64
wt      float64
qsec      float64
vs      int64
am      int64
gear      int64
carb      int64
dtype: object

```

```

# Compute and print the matrix
matrix = df.corr(numeric_only=True)
print(matrix)

```

```

      S.No      mpg      cyl      disp      hp      drat      wt  \
S.No  1.000000  0.084699 -0.028882 -0.022048  0.242295  0.240952 -0.162419
mpg   0.084699  1.000000 -0.852162 -0.847551 -0.776168  0.681172 -0.867659
cyl   -0.028882 -0.852162  1.000000  0.902033  0.832447 -0.699938  0.782496
disp  -0.022048 -0.847551  0.902033  1.000000  0.790949 -0.710214  0.887980
hp     0.242295 -0.776168  0.832447  0.790949  1.000000 -0.448759  0.658748
drat   0.240952  0.681172 -0.699938 -0.710214 -0.448759  1.000000 -0.712441
wt     -0.162419 -0.867659  0.782496  0.887980  0.658748 -0.712441  1.000000
qsec  -0.362390  0.360351  0.548775 -0.385207 -0.650674  0.120175 -0.130362
vs     -0.109162  0.664039 -0.810812 -0.710416 -0.723097  0.440278 -0.554916
am     0.354905  0.599832 -0.522607 -0.591227 -0.243204  0.712711 -0.692495
gear   0.400825  0.480285 -0.492687 -0.555569 -0.125704  0.699610 -0.583287
carb   0.174576 -0.550925  0.526988  0.394977  0.749812 -0.090790  0.427606

      qsec      vs      am      gear      carb
S.No -0.362390 -0.109162  0.354905  0.400825  0.174576
mpg   0.360351  0.664039  0.599832  0.480285 -0.550925
cyl   -0.548775 -0.810812 -0.522607 -0.492687  0.526988
disp  -0.385207 -0.710416 -0.591227 -0.555569  0.394977
hp     -0.650674 -0.723097 -0.243204 -0.125704  0.749812
drat   0.120175  0.440278  0.712711  0.699610 -0.090790
wt     -0.130362 -0.554916 -0.692495 -0.583287  0.427606

```

```
qsec 1.000000 0.667873 -0.271763 -0.203784 -0.573987
vs 0.667873 1.000000 0.168345 0.206023 -0.569607
am -0.271763 0.168345 1.000000 0.794059 0.057534
gear -0.203784 0.206023 0.794059 1.000000 0.274073
carb -0.573987 -0.569607 0.057534 0.274073 1.000000
```

```
# View hp column as a Series
print(df['hp'])
```

```
0    110
1    110
2     93
3    110
4    175
5    105
6    245
7     62
8     95
9    123
10   123
11   180
12   180
13   180
14   205
15   215
16   230
17     66
18     52
19     65
20     97
21   150
22   150
23   245
24   175
25     66
26     91
27   113
28   264
29   175
30   335
31   109
Name: hp, dtype: int64
```

```
df['hp'].head()
```

```
   hp
0  110
1  110
2   93
3  110
4  175

dtype: int64
```

```
# see all rows and all columns
print(df.loc[:, :])
```

	S.No	model	mpg	cyl	disp	hp	drat	wt	qsec	\
0	1	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.460000	
1	2	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.020000	
2	3	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.610000	
3	4	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.440000	
4	5	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.020000	
5	6	Valiant	18.1	6	225.0	105	2.76	3.460	17.674828	
6	7	Duster 360	14.3	8	360.0	245	3.21	3.570	15.840000	
7	8	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.000000	
8	9	Merc 230	22.8	4	140.8	95	3.92	3.150	22.900000	
9	10	Merc 280	19.2	6	167.6	123	3.92	3.440	18.300000	
10	11	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.900000	
11	12	Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.400000	
12	13	Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.600000	
13	14	Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.000000	
14	15	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.980000	
15	16	Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.820000	

16	17	Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.420000
17	18	Fiat 128	32.4	4	78.7	66	4.08	2.200	17.674828
18	19	Honda Civic	30.4	4	75.7	52	4.93	1.615	18.520000
19	20	Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.900000
20	21	Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.010000
21	22	Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.870000
22	23	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.300000
23	24	Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.410000
24	25	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.050000
25	26	Fiat X1-9	27.3	4	79.0	66	4.08	1.935	17.674828
26	27	Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.700000
27	28	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.900000
28	29	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.500000
29	30	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.500000
30	31	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.600000
31	32	Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.600000

	vs	am	gear	carb
0	0	1	4	4
1	0	1	4	4
2	1	1	4	1
3	1	0	3	1
4	0	0	3	2
5	1	0	3	1
6	0	0	3	4
7	1	0	4	2
8	1	0	4	2
9	1	0	4	4
10	1	0	4	4
11	0	0	3	3
12	0	0	3	3
13	0	0	3	3
14	0	0	3	4
15	0	0	3	4
16	0	0	3	4
17	1	1	4	1
18	1	1	4	2
19	1	1	4	1
20	1	0	3	1
21	0	0	3	2

```
# Select rows from index 6 to the end, and columns from 'hp' to 'carb'
```

```
subset_df = df.loc[6:, 'hp':'carb']
print(subset_df.to_string())
```

	hp	drat	wt	qsec	vs	am	gear	carb
6	245	3.21	3.570	15.840000	0	0	3	4
7	62	3.69	3.190	20.000000	1	0	4	2
8	95	3.92	3.150	22.900000	1	0	4	2
9	123	3.92	3.440	18.300000	1	0	4	4
10	123	3.92	3.440	18.900000	1	0	4	4
11	180	3.07	4.070	17.400000	0	0	3	3
12	180	3.07	3.730	17.600000	0	0	3	3
13	180	3.07	3.780	18.000000	0	0	3	3
14	205	2.93	5.250	17.980000	0	0	3	4
15	215	3.00	5.424	17.820000	0	0	3	4
16	230	3.23	5.345	17.420000	0	0	3	4
17	66	4.08	2.200	17.674828	1	1	4	1
18	52	4.93	1.615	18.520000	1	1	4	2
19	65	4.22	1.835	19.900000	1	1	4	1
20	97	3.70	2.465	20.010000	1	0	3	1
21	150	2.76	3.520	16.870000	0	0	3	2
22	150	3.15	3.435	17.300000	0	0	3	2
23	245	3.73	3.840	15.410000	0	0	3	4
24	175	3.08	3.845	17.050000	0	0	3	2
25	66	4.08	1.935	17.674828	1	1	4	1
26	91	4.43	2.140	16.700000	0	1	5	2
27	113	3.77	1.513	16.900000	1	1	5	2
28	264	4.22	3.170	14.500000	0	1	5	4
29	175	3.62	2.770	15.500000	0	1	5	6
30	335	3.54	3.570	14.600000	0	1	5	8
31	109	4.11	2.780	18.600000	1	1	4	2

```
# Select all rows, index 0 (the first column)
```

```
first_column = df.iloc[:, 0]
print(first_column.to_string())
```

```

0      1
1      2
2      3
3      4
4      5
5      6
6      7
7      8
8      9
9     10
10     11
11     12
12     13
13     14
14     15
15     16
16     17
17     18
18     19
19     20
20     21
21     22
22     23
23     24
24     25
25     26
26     27
27     28
28     29
29     30
30     31
31     32

```

```

# Select all rows using ':', and select the 'mpg' column by its label name

mpg_records = df.loc[:, 'mpg']#OR = print(df.loc[:, ['mpg']].to_string())

# Print every single record without truncation
print(mpg_records.to_string())

```

```

0      21.0
1      21.0
2      22.8
3      21.4
4      18.7
5      18.1
6      14.3
7      24.4
8      22.8
9      19.2
10     17.8
11     16.4
12     17.3
13     15.2
14     10.4
15     10.4
16     14.7
17     32.4
18     30.4
19     33.9
20     21.5
21     15.5
22     15.2
23     13.3
24     19.2
25     27.3
26     26.0
27     30.4
28     15.8
29     19.7
30     15.0
31     21.4

```

```

#display the records from index 0 to index 6 from mpg column

print(df.loc[0:6, ['mpg']])

```

```

      mpg
0  21.0
1  21.0

```

```

2  22.8
3  21.4
4  18.7
5  18.1
6  14.3

```

```
print(df.loc[:7, ['mpg']])
```

```

      mpg
0  21.0
1  21.0
2  22.8
3  21.4
4  18.7
5  18.1
6  14.3
7  24.4

```

```

# Set all values in the 'am' column to 1
df['am'] = 1

# View the first few rows to verify the change
print(df[['model', 'am']].head())

```

```

      model  am
0    Mazda RX4    1
1  Mazda RX4 Wag    1
2   Datsun 710    1
3  Hornet 4 Drive    1
4  Hornet Sportabout    1

```

```

# Double each value in the 'am' column using lambda

df['am'] = df['am'].apply(lambda x: x * 2)
print(df[['am']])

```

```

      am
2      8
7      8
8      8
31     8
20     8
19     8
18     8
17     8
25     8
26     8
27     8
1      8
3      8
5      8
9      8
0      8
29     8
10     8
13     8
14     8
11     8
6      8
4      8
12     8
15     8
22     8
21     8
16     8
23     8
24     8
28     8
30     8

```

```

# Sort by cyl in ascending order

df = df.sort_values(by='cyl', ascending=True)
print(df[['cyl']])

```

```

      cyl
2       4

```

```
7    4
8    4
31   4
20   4
19   4
18   4
17   4
25   4
26   4
27   4
1    6
3    6
5    6
9    6
0    6
29   6
10   6
13   8
14   8
11   8
6    8
4    8
12   8
15   8
22   8
21   8
16   8
23   8
24   8
28   8
30   8
```

```
df=df.sort_values(by='cyl',ascending=False)
df[['cyl']]
```

	cyl
15	8
22	8
30	8
28	8
16	8
21	8
24	8
23	8
11	8
6	8
12	8
4	8
13	8
14	8
3	6
5	6
0	6
9	6
10	6
1	6
29	6
26	4
27	4
25	4
7	4
8	4
2	4
31	4
20	4
19	4
18	4

```
df[df['cyl']>6]
```

	S.No	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
15	16	Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	8	3	4
22	23	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	8	3	2
30	31	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	8	5	8
28	29	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	8	5	4
16	17	Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	8	3	4
21	22	Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	8	3	2
24	25	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	8	3	2
23	24	Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	8	3	4
11	12	Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	8	3	3
6	7	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	8	3	4
12	13	Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	8	3	3
4	5	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	8	3	2
13	14	Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	8	3	3
14	15	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	8	3	4

```
cyl1300 = df[(df['cyl']>6) & (df['hp']>300) ]
print(cyl1300)
```

```

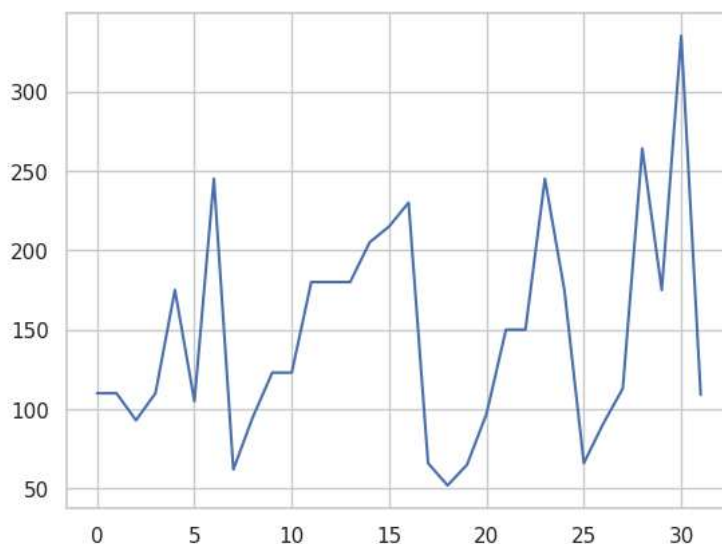
      S.No      model  mpg  cyl  disp  hp  drat   wt   qsec  vs  am  \
30      31  Maserati Bora  15.0    8  301.0  335  3.54  3.57  14.6  0   8
      gear  carb
30      5     8
```

```
# 1. Reload your original dataset to restore all 32 rows
# (Replace 'your_data.csv' with your actual file path)
df_original = pd.read_csv('/content/mtcars2.csv')

# 2. Make sure the horsepower column is a float type
df_original['hp'] = pd.to_numeric(df_original['hp'], errors='coerce')

# 3. Generate the exact line plot using the full, unfiltered column
plt.plot(df_original['hp'])

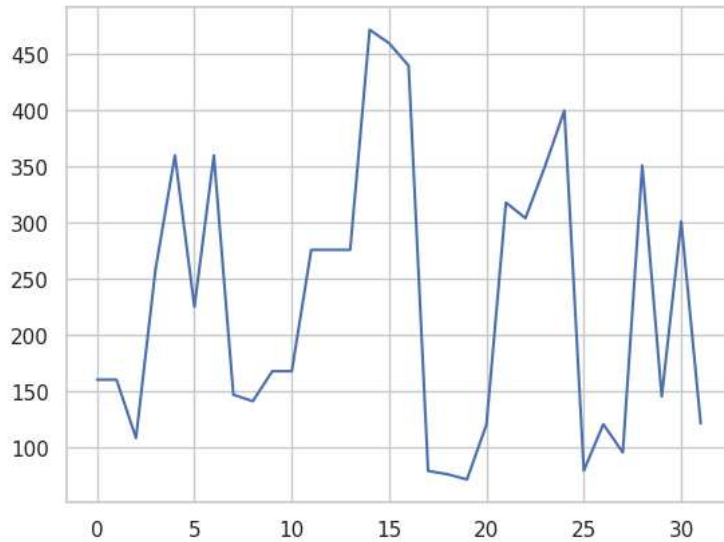
# 4. Display the plot window
plt.show()
```



```
# Generate the exact line plot for the displacement column
plt.plot(df_original['disp'])
```



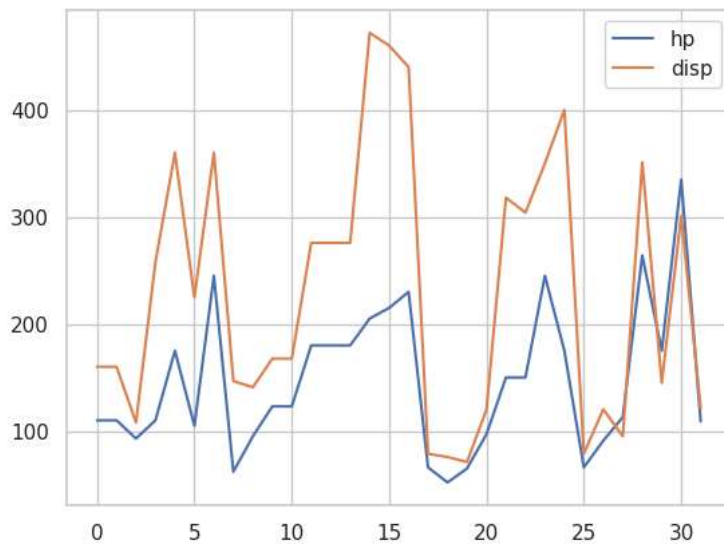
```
# Force the visual window to display on screen
plt.show()
```



```
# Plot both columns at the same time
plt.plot(df_original[['hp', 'disp']])

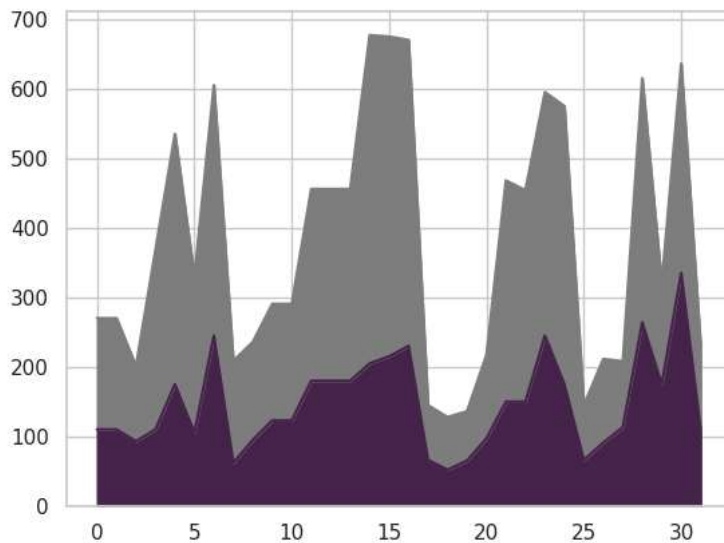
# Add the legend to match the top-right label box in your image
plt.legend(['hp', 'disp'])

# Display the final plot window
plt.show()
```



```
# Create the stacked area plot with specific colors matching your image
# 'color' accepts hex codes or named colors; 'legend=False' hides the label box
df_original[['hp', 'disp']].plot.area(color=['#4A234F', '#7F7F7F'], legend=False)

# Display the final plot window
plt.show()
```

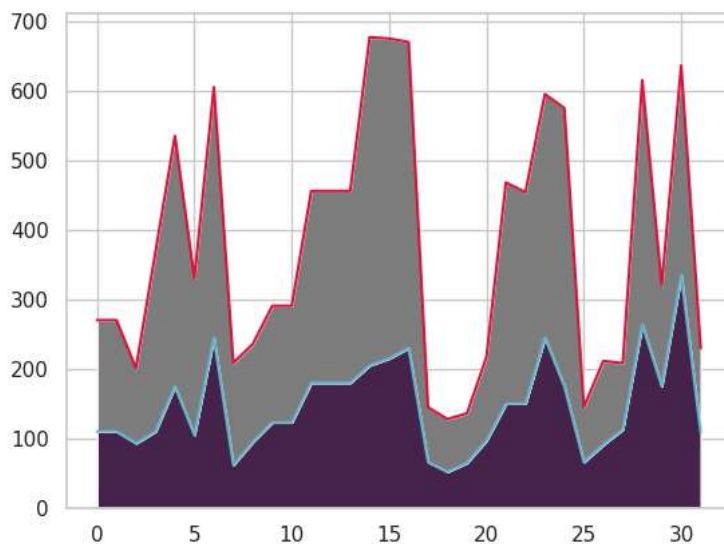


```
# 1. Plot the filled area stack (Dark Purple at bottom, Gray on top)
# Using stackplot calculates the stacked heights automatically
plt.stackplot(df_original.index, df_original['hp'], df_original['disp'],
              colors=['#4A234F', '#7F7F7F'])

# 2. Add the cyan stroke for the 'hp' line (the lower boundary line)
plt.plot(df_original['hp'], color='c', linewidth=1.5)

# 3. Add the red stroke for the stacked 'hp + disp' line (the upper boundary line)
# Because stackplot accumulates columns, the top line tracks the sum of both values
plt.plot(df_original['hp'] + df_original['disp'], color='crimson', linewidth=1.5)

# 4. Display the clean visual layout
plt.show()
```



```
# 1. model to list
# (Ensures categorical text labels are handled neatly as a list sequence)
car_models = df_original['Unnamed: 1'].tolist()
hp_values = df_original['hp'].tolist()

# 2. adding figure to adjust figsize
# (An extra-wide layout ensures all 32 labels fit side-by-side)
plt.figure(figsize=(20, 10))

# 3. see how hp changes with bar plot
# (Plots the bar graph using a solid purple color)
plt.bar(car_models, hp_values, color='purple')

# Adjust layout boundaries and display the plot window
```

```
plt.tight_layout()
```